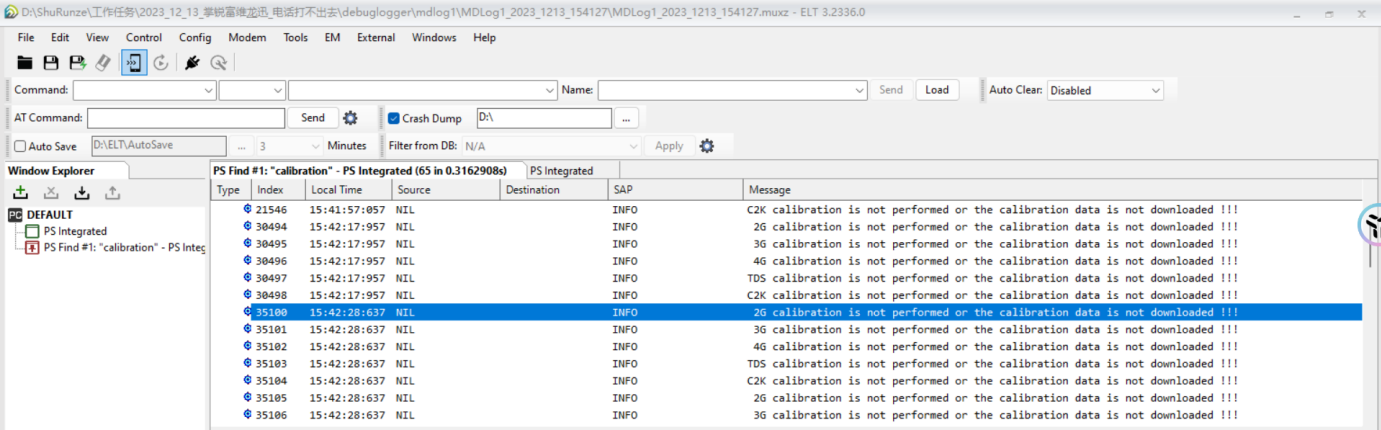


modem log分析方法

1、查看是否校准

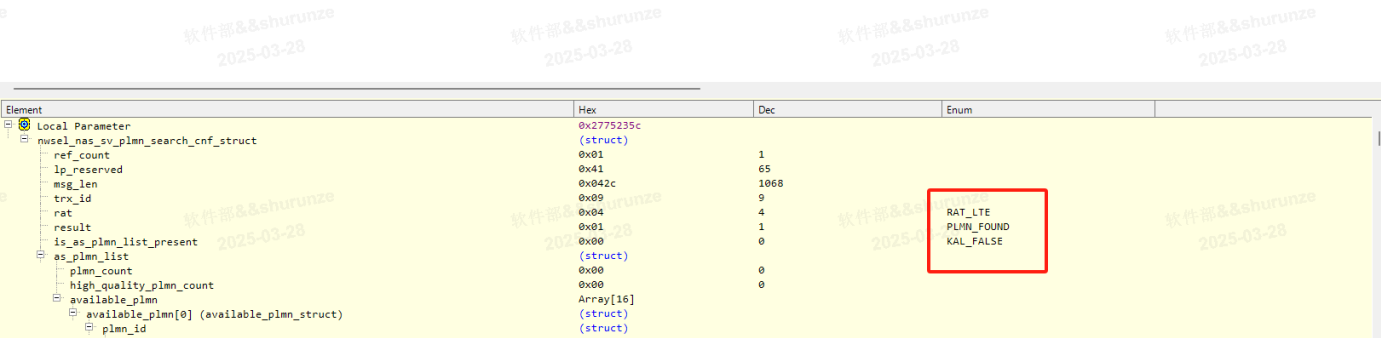
搜索calibration



2、查看搜网驻网情况

搜索MSG_ID_NWSEL_NAS_SV_PLMN_SEARCH_CNF检查搜网情况

Type	Index	No	Frame#	Time	Local Time	Source	Destination	SAP	Message
8526	0	0	88723	15:20:35:493	NAS_SV	NWSEL	NAS_SV_SAP		MSG_ID_NWSEL_NAS_SV_PLMN_SEARCH_CNF
94944	1	0	113889	15:20:37:695	INVALID	INVALID			Discard Info: PS_TRACE: 238 PS_PROHIBITIVE: 44
243129	2	1738171	7400839	15:28:23:973	NAS_SV	NWSEL	NAS_SV_SAP		MSG_ID_NWSEL_NAS_SV_PLMN_SEARCH_CNF
380258	3	1766002	7743282	15:28:45:729	NAS_SV	NWSEL	NAS_SV_SAP		MSG_ID_NWSEL_NAS_SV_PLMN_SEARCH_CNF
805206	4	1619199	8133641	15:29:10:216	NAS_SV	NWSEL	NAS_SV_SAP		MSG_ID_NWSEL_NAS_SV_PLMN_SEARCH_CNF
805940	5	1619252	8133694	15:29:10:216	NAS_SV	NWSEL	NAS_SV_SAP		MSG_ID_NWSEL_NAS_SV_PLMN_SEARCH_CNF
822582	6	2408891	8210637	15:29:15:800	NAS_SV	NWSEL	NAS_SV_SAP		MSG_ID_NWSEL_NAS_SV_PLMN_SEARCH_CNF
1045238	7	1640452	8968268	15:30:04:279	NAS_SV	NWSEL	NAS_SV_SAP		MSG_ID_NWSEL_NAS_SV_PLMN_SEARCH_CNF



表明搜索到了4G的PLMN，即将开始驻网

查看OTA message对应时间点显示的驻网情况

Type	Index	Local Time	Time D...	AP UTC Time	Direction	Layer	Message Type	Note
⊕	378806	15:28:45:729		15:28:45.563056	NW->MS	ERRC	SystemInformation	(EARFCN[38400], PCI[191])
⊕	378807	15:28:45:729		15:28:45.563056	NW->MS	ERRC	SystemInformationBlockType2	(EARFCN[38400], PCI[191])
⊕	378808	15:28:45:729		15:28:45.563056	NW->MS	ERRC	SystemInformationBlockType3	(EARFCN[38400], PCI[191])
⊕	385042	15:28:45:929		15:28:45.603056	NW->MS	ERRC	SystemInformation	(EARFCN[38400], PCI[191])
⊕	385043	15:28:45:929		15:28:45.603056	NW->MS	ERRC	SystemInformationBlockType5	(EARFCN[38400], PCI[191])
⊕	387675	15:28:45:929		15:28:45.619312	MS->NW	ESM	MSG_PDN_CONNECTIVITY_REQUEST	(PTI:13, EBI:0)
⊕	389285	15:28:45:929		15:28:45.644912	MS->NW	ERRC	RRCConnectionRequest	(EARFCN[38400], PCI[191])
⊕	389696	15:28:45:929		15:28:45.681328	NW->MS	ERRC	RRCConnectionSetup	(EARFCN[38400], PCI[191])
⊕	391071	15:28:45:929		15:28:45.694832	MS->NW	EMM	Attach_Request	(EPS attach type="EMM_ATTACH_TYPE_COMBINED_ATTACH")
⊕	391485	15:28:45:929		15:28:45.697008	MS->NW	ERRC	RRCConnectionSetupComplete	(EARFCN[38400], PCI[191])
⊕	392452	15:28:45:929		15:28:45.752048	NW->MS	ERRC	DLInformationTransfer	(EARFCN[38400], PCI[191])
⊕	392515	15:28:45:929		15:28:45.752048	NW->MS	EMM	Identity_Request	(Identity type="D_EMM_ID_TYPE_INEISV")
⊕	392628	15:28:45:929		15:28:45.753264	MS->NW	EMM	Identity_Response	
⊕	392679	15:28:45:929		15:28:45.753584	MS->NW	ERRC	ULInformationTransfer	(EARFCN[38400], PCI[191])
⊕	392903	15:28:45:929		15:28:45.772080	NW->MS	ERRC	DLInformationTransfer	(EARFCN[38400], PCI[191])
⊕	392997	15:28:45:929		15:28:45.772592	NW->MS	ESM	MSG_ESM_INFORMATION_REQUEST	(PTI:13, EBI:0)
⊕	393154	15:28:45:929		15:28:45.773808	MS->NW	ESM	MSG_ESM_INFORMATION_RESPONSE	(PTI:13, EBI:0)
⊕	393353	15:28:45:929		15:28:45.775216	MS->NW	ERRC	ULInformationTransfer	(EARFCN[38400], PCI[191])
⊕	395663	15:28:46:330		15:28:46.002032	NW->MS	ERRC	UECapabilityEnquiry	(EARFCN[38400], PCI[191])(EUTRA[1], UTRA[1], GERAN-CS[1], GERAN-PS[1], C2K-1xRTT[1])

Element	Hex	Dec	Enum
[MS->NW] EMM_Attach_Request(EPS attach type="EMM_ATTACH_TYPE_COMBINED_ATTACH")			
global_peer_msg_id	0x0000 0321	801	PROTOCOL_SECTION__NAS_EPS_PLAIN
raw data			
0000 - 000F	07 41 22 0b f6 64 f0 00	08 a0 22 d9 82 3f 2b 07	
0010 - 001F	f0 f0 00 80 1c 80 00 00	05 02 0d d0 31 d1 52 64	
0020 - 002F	f0 00 33 13 13 64 f0 00	33 0f 11 03 4f 18 b0 f1	
0030 - 003F	5d 01 03 e0 c1 10 02 f0	c0	
Message Content			
Non-Access-Stratum (NAS)PDU			
0000 = Security header type: Plain NAS message, not security protected (0)			
.... 0111 = Protocol discriminator: EPS mobility management messages (0x7)			
NAS EPS Mobility Management Message Type: Attach request (0x41)			
0 = Type of security context flag (TSC): Native security context (for KSIaue or KSIauef)			
..010 = NAS key set identifier: (2)			
.... 0 ... = Spare bit(s): 0x00			
.... 010 = EPS attach type: Combined EPS/IMS/I attach (2)			

可以看到发起的驻网请求类型是NAS EPS，说明发起的是驻4G网的请求

Type	Index	Local Time	Time D...	AP UTC Time	Direction	Layer	Message Type	Note
⊕	392628	15:28:45:929		15:28:45.753264	MS->NW	EMM	Identity_Response	
⊕	392679	15:28:45:929		15:28:45.753584	MS->NW	ERRC	ULInformationTransfer	(EARFCN[38400], PCI[191])
⊕	392903	15:28:45:929		15:28:45.772080	NW->MS	ERRC	DLInformationTransfer	(EARFCN[38400], PCI[191])
⊕	392997	15:28:45:929		15:28:45.772592	NW->MS	ESM	MSG_ESM_INFORMATION_REQUEST	(PTI:13, EBI:0)
⊕	393154	15:28:45:929		15:28:45.773808	MS->NW	ESM	MSG_ESM_INFORMATION_RESPONSE	(PTI:13, EBI:0)
⊕	393353	15:28:45:929		15:28:45.775216	MS->NW	ERRC	ULInformationTransfer	(EARFCN[38400], PCI[191])
⊕	395663	15:28:46:330		15:28:46.002032	NW->MS	ERRC	UECapabilityEnquiry	(EARFCN[38400], PCI[191])(EUTRA[1], UTRA[1], GERAN-CS[1], GERAN-PS[1], C2K-1xRTT[1])
⊕	395899	15:28:46:330		15:28:46.003504	MS->NW	ERRC	UE-EUTRA-capability	(AS-Rel15, ue-CategoryDL[18], ue-CategoryUL[18], EN-DC[KAL_FALSE])
⊕	395932	15:28:46:330		15:28:46.003760	MS->NW	ERRC	UECapabilityInformation	(EARFCN[38400], PCI[191])(EUTRA[1], UTRA[0], GERAN-CS[0], GERAN-PS[0], C2K-1xRTT[0])
⊕	396316	15:28:46:330		15:28:46.024176	NW->MS	ERRC	SecurityModeCommand	(EARFCN[38400], PCI[191])(Cipher Algo:[CipheringAlgorithm_r12_eea2], Integrity Algo:[CipheringAlgorithm_r12_integ])
⊕	396409	15:28:46:330		15:28:46.025040	MS->NW	ERRC	SecurityModeComplete	(EARFCN[38400], PCI[191])
⊕	396595	15:28:46:330		15:28:46.041136	NW->MS	ERRC	RRCConnectionReconfiguration	(EARFCN[38400], PCI[191])(measCfg:[0], mobCtrlInfo:[0], dedInfoNASList:[1], radioresCfg:[0])
⊕	397751	15:28:46:330		15:28:46.051088	MS->NW	ERRC	RRCConnectionReconfigurationComplete	(EARFCN[38400], PCI[191])
⊕	397905	15:28:46:330		15:28:46.052336	NW->MS	EMM	Attach_Accept	(EPS attach result="EMM_ATTACH_RESULT_COMBINED_ATTACHED")
⊕	398463	15:28:46:330		15:28:46.054832	NW->MS	ESM	MSG_ACTIVATE_DEFAULT_EPS_BEARER_CONTEXT_REQUEST	(PTI:13, EBI:5)
⊕	399179	15:28:46:330		15:28:46.057264	MS->NW	ESM	MSG_ACTIVATE_DEFAULT_EPS_BEARER_CONTEXT_ACCEPT	(PTI:0, EBI:5)
⊕	399689	15:28:46:330		15:28:46.061424	MS->NW	EMM	Attach_Complete	
⊕	399842	15:28:46:330		15:28:46.062128	MS->NW	ERRC	ULInformationTransfer	(EARFCN[38400], PCI[191])
⊕	400102	15:28:46:330		15:28:46.067312	NW->MS	ERRC	RRCConnectionReconfiguration	(EARFCN[38400], PCI[191])(measCfg:[1], mobCtrlInfo:[0], dedInfoNASList:[0], radioresCfg:[0])

Element	Hex	Dec	Enum
[NW->MS] EMM_Attach_Accept(EPS attach result="EMM_ATTACH_RESULT_COMBINED_ATTACHED")			
global_peer_msg_id	0x0000 0321	801	PROTOCOL_SECTION__NAS_EPS_PLAIN
raw data			
0000 - 000F	07 42 02 49 06 20 64 f0	00 33 13 00 5f 52 0d c1	
0010 - 001F	01 09 23 0c 63 6d 69 6f	76 74 63 6d 61 63 6c 77	
0020 - 002F	02 6a 73 06 6d 6e 63 30	31 33 06 6d 63 63 34 36	
0030 - 003F	30 04 67 70 72 73 05 01	0a 26 11 1d 5e 04 fe fe	
0040 - 004F	a3 a3 58 32 27 26 80 80	21 0a 03 01 00 0a 81 06	
0050 - 005F	ac 1d 01 c8 80 21 0a 04	01 00 0a 83 06 00 00 00	
0060 - 006F	00 00 0d 04 ac 1d 01 c8	00 05 01 02 50 0b f6 64	
0070 - 007F	f0 00 08 a0 22 e9 82 3f	2b 13 64 f0 00 33 0f 17	
0080 - 008F	2c 59 2c 64 01 05 f2		
Message Content			
Non-Access-Stratum (NAS)PDU			
0000 = Security header type: Plain NAS message, not security protected (0)			
.... 0111 = Protocol discriminator: EPS mobility management messages (0x7)			
NAS EPS Mobility Management Message Type: Attach accept (0x42)			
0000 = Spare half octet: 0			
.... 0 ... = Spare bit(s): 0x00			

驻网被接受，驻网成功

当然也有驻网被拒的情况，下面被拒原因是 ESM_FAILURE，一般是无流量导致

PS Find #1: "MSG_ID_NWSEL...MN_SEARCH_CNF" - PS Integrated (4 in 4.97014s)					OTA Messages		PS Integrated	System Trace	PS Find #2: "EOPS" - System Trace (476 in 0.0758533s)	
pe	Index	Local Time	Time D...	AP UTC Time	Direction	Layer	Message Type		Note	
	156265	09:09:42:018		09:09:42.152000	MS->NW	EMM	Security_Mode_Complete			
	156368	09:09:42:018		09:09:42.152704	MS->NW	ERRC	ULInformationTransfer		(EARFCN[6200], PCI[225])	
	157386	09:09:42:219		09:09:42.319552	NW->MS	ERRC	DLInformationTransfer		(EARFCN[6200], PCI[225])	
	157473	09:09:42:219		09:09:42.320064	NW->MS	ESM	MSG_ESM_INFORMATION_REQUEST		(PTI:3, EBI:0)	
	157556	09:09:42:219		09:09:42.320960	MS->NW	ESM	MSG_ESM_INFORMATION_RESPONSE		(PTI:3, EBI:0)	
	157753	09:09:42:219		09:09:42.322112	MS->NW	ERRC	ULInformationTransfer		(EARFCN[6200], PCI[225])	
	158074	09:09:42:219		09:09:42.342592	NW->MS	ERRC	DLInformationTransfer		(EARFCN[6200], PCI[225])	
	158141	09:09:42:219		09:09:42.342912	NW->MS	EMM	Attach_Reject		(EMM cause="EMM_CAUSE_ESM_FAILURE")	
	158168	09:09:42:219		09:09:42.343424	NW->MS	ERRC	RRCConnectionRelease		(EARFCN[6200], PCI[225])(cause:[Release	
	158398	09:09:42:219		09:09:42.344576	NW->MS	ESM	MSG_PDN_CONNECTIVITY_REJECT		(PTI:3, EBI:0, ESM cause:"ESM_CAUSE_UNK	
	162136	09:09:42:219		09:09:42.380352	NW->MS	ERRC	MasterInformationBlock		(EARFCN[6200], PCI[225])	
	162771	09:09:42:219		09:09:42.396608	NW->MS	ERRC	SystemInformationBlockType1		(EARFCN[6200], PCI[225])	
	169865	09:09:43:839		09:09:43.890432	NW->MS	ERRC	MasterInformationBlock		(EARFCN[37950], PCI[466])	

Element		Hex	Dec	Enum
raw data		Buffer		
0000 - 000F		07 44 13 78 00 29 02 03	d1 1b 27 23 80 80 21 10	
0010 - 001F		01 01 00 10 81 06 00 00	00 00 83 06 00 00 00 00	
0020 - 002E		00 0a 00 00 0d 00 00 05	00 00 11 00 00 10 00	
Message Content				
Non-Access-Stratum (NAS)PDU				
0000 = Security header type: Plain NAS message, not security protected (0)				
.... 0111 = Protocol discriminator: EPS mobility management messages (0x7)				
NAS EPS Mobility Management Message Type: Attach reject (0x44)				
EMM cause				
Cause: ESM failure (19)				
ESM message container				
Element ID: 0x78				
Length: 41				
ESM message container contents:				

3、查看选择的小区

搜索cell select:可以看到所选择的小区的PCI（物理小区标识）和earfcn（E-UTRA绝对无线频率信道号）

Type	Index	No	Frame #	Time	Local Time	Source	SAP	Message	
	35790	0	1723497	45444987	12:11:38:893	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[41261]/pci[247]	
	74148	1	1811165	45532655	12:11:44:503	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[1345]/pci[294]	
	86054	2	1835610	45557100	12:11:46:107	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[1254]/pci[154]	
	98790	3	1837784	45583724	12:11:47:710	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[1899]/pci[499]	
	133461	4	1889485	45662062	12:11:52:713	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[38407]/pci[405]	
	137942	5	1896427	45669004	12:11:53:117	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[38319]/pci[501]	
	169285	6	1862625	45720467	12:11:56:518	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[1552]/pci[502]	
	172126	7	1812733	45722034	12:11:56:518	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[1552]/pci[502]	
	177639	8	1816859	45727721	12:11:56:918	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[1552]/pci[502]	
	182775	9	1815187	45731737	12:11:57:118	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[1552]/pci[502]	
	187407	10	1820296	45736846	12:11:57:518	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[1552]/pci[502]	
	327879	11	1841154	46206782	12:12:27:525	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[1552]/pci[502]	
	355500	12	1902889	46268517	12:12:31:530	ERR...	ERRC_CEL_BASELINE_H	[CEL_DI/LV] cell select: earfcn[9452]/pci[228]	
Element						Hex		Dec	Enum
[CEL_DI/LV] cell select: earfcn[9386]/pci[132]									
Trace Id: 741343368									
Trace Name: ERRC_CEL_ISCELLS_DI_SELECT_CELL									

4、查看天线接收信号情况

搜索RSRP(RX0, RX1)，RX0表示主天线的信号，RX1表示副天线的信号

PS Integrated	PS Find #9: "...	OTA Messag...	System Trace	PS Find #1: "...	PS Find #2: "...	PS Find #3: "...	PS Find #4: "...	PS Find #5: "...	PS Find #6: "...	PS Find #7: ...	PS Find #8: "...	PS Find #1...
Local Time	Source	SAP	Message									
12:13:17:561	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7756, PCI = 132, RSRP(RX0, RX1) = (-150, -150) dBm, R									
12:13:17:561	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7756, PCI = 132, RSRP(RX0, RX1) = (-150, -112) dBm, R									
12:13:17:561	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7756, PCI = 132, RSRP(RX0, RX1) = (-110, -130) dBm, R									
12:13:17:561	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7756, PCI = 132, RSRP(RX0, RX1) = (-113, -115) dBm, R									
12:13:17:561	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7756, PCI = 132, RSRP(RX0, RX1) = (-113, -150) dBm, R									
12:13:17:561	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7756, PCI = 132, RSRP(RX0, RX1) = (-150, -120) dBm, R									
12:13:17:561	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7756, PCI = 132, RSRP(RX0, RX1) = (-150, -150) dBm, R									
12:13:22:968	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7640, PCI = 36, RSRP(RX0, RX1) = (-105, -116) dBm, RS									
12:13:22:968	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7640, PCI = 36, RSRP(RX0, RX1) = (-103, -117) dBm, RS									
12:13:22:968	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7640, PCI = 36, RSRP(RX0, RX1) = (-119, -150) dBm, RS									
12:13:22:968	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7640, PCI = 36, RSRP(RX0, RX1) = (-150, -150) dBm, RS									
12:13:22:968	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7640, PCI = 36, RSRP(RX0, RX1) = (-114, -118) dBm, RS									
12:13:22:968	EL1...	EL1_MPC_BASELINE_PUBLIC_UH	[EL1MPC_MEAS][L1F] L1 filtering results: Band = LTE_BAND28, DL_Freq = 7640, PCI = 36, RSRP(RX0, RX1) = (-150, -150) dBm, RS									
Element	Hex			Dec			Enum					

4G 信号阈值如下:

rsrp	< -128	-128 ~ -118	-118 ~ -108	-108 ~ -98	> -98
Level	0	1	2	3	4

5、查看当前设置的网络制式

搜索ERAT，6表示当前设置的网络制式是GSM+UMTS+LTE，4表示LTE优先

ocal Time	Source	Destination	SAP	Message
2:13:22:968	NIL		FUNC	[ATP_AT_R_0,ch2, s00002ebb]+ERAT: 255,0,6,4,0
2:13:22:968	NIL		FUNC	[ATP_AT_I_0,ch2, s00002ebc]+ERAT?
2:13:22:968	NIL		FUNC	[ATCI_AT_I_0 s7]+ERAT?
2:13:22:968	NIL		FUNC	[ATCI_AT_R_0 s7]+ERAT: 255,0,6,4,0
2:13:22:968	NIL		FUNC	[ATP_AT_R_0,ch2, s00002ebc]+ERAT: 255,0,6,4,0
2:13:23:371	NIL		FUNC	[ATP_AT_I_0,ch2, s00002ec0]+ERAT?
2:13:23:371	NIL		FUNC	[ATCI_AT_I_0 s7]+ERAT?
2:13:23:371	NIL		FUNC	[ATCI_AT_R_0 s7]+ERAT: 255,0,6,4,0
2:13:23:371	NIL		FUNC	[ATP_AT_R_0,ch2, s00002ec0]+ERAT: 255,0,6,4,0
2:13:23:371	NIL		FUNC	[ATP_AT_I_0,ch2, s00002ec1]+ERAT?
2:13:23:371	NIL		FUNC	[ATCI_AT_I_0 s7]+ERAT?
2:13:23:371	NIL		FUNC	[ATCI_AT_R_0 s7]+ERAT: 255,0,6,4,0
2:13:23:371	NIL		FUNC	[ATP_AT_R_0,ch2, s00002ec1]+ERAT: 255,0,6,4,0

Format

Command	Possible Response(s)
+ERAT?	+ERAT : <Act>, <GPRS status>,<RAT mode>,<prefer_rat>,
+ERAT=<RAT mode>[,<prefer_rat> [,<lock>][,<is_not_saved>]]	OK /ERROR

<RAT mode>: RAT mode setting of MS

0: GSM only

1: UMTS only

2: GSM + UMTS

3: LTE only

4: GSM + LTE

5: UMTS + LTE

6: GSM + UMTS + LTE

7: C2K only

8: GSM + C2K

<prefer_rat>: the field is discarded for 3GPP RAT, please use AT+EPRATL to set prefer RAT list for 3GPP RAT

0: No prefer

1: GSM prefer

2: UMTS prefer

4: LTE prefer

32: 3GPP2 1xRTT

64: 3GPP2 HRPD

128: NR prefer

6、查看当前驻留小区的运营商代码以及网络制式

搜索EOPS，46001是当前驻留小区的mccmnc，1表示当前网络制式是GPRS

Local Time	Source	Destination	SAP	Message
12:11:57:718	NIL		FUNC	[ATCI_AT_R_0 s61]+EOPS: 0,2,"46001",1
12:11:57:718	NIL		FUNC	[ATP_AT_R_0,ch2, s00002c14]+EOPS: 0,2,"46001",1
12:11:57:718	NIL		FUNC	[ATCI_AT_R_0 s61]+EOPS: 0,2,"46001",1
12:11:57:718	NIL		FUNC	[ATP_AT_R_0,ch2, s00002c20]+EOPS: 0,2,"46001",1

+EOPS?	+EOPS: <mode>,<format>,<oper>,<act> +CME ERROR: <err>
--------	---

<eAcT>: access technology selected

0x0001 L4C_GPRS_SUPPORT

0x0002 L4C_EDGE_SUPPORT

0x0004 L4C_UMTS_SUPPORT

0x0008 L4C_HSDPA_SUPPORT

0x0010 L4C_HSUPA_SUPPORT

0x0018 L4C_HSDPA_UPA_SUPPORT

0x1000 L4C_LTE_SUPPORT

0x0100 L4C_1XRTT_SUPPORT

0x0200 L4C_HRPD_SUPPORT

0x0400 L4C_EHRPD_SUPPORT

7、查看驻网的band

搜索MOD_MRS

PS Integrated		OTA Messages	System Trace	PS Find #1: "LTE_UL_MCS" - PS Integrated (...)			PS Find #2: "mcs" - PS Integrated (0 in 4,765...		PS Find #3: "sinr" - PS Integrated (4223 in 4...		PS Find #4: "band" - PS Find #3 (3526 in 0.1...		PS Find #5: "MOD_MRS" - PS Integrated ...	
Type	Index	No	Frame #	Time	Local Time	Source	Destination	SAP	Message					
	98202	236	1215166	1103989478	12:39:10:127	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98203	237	1215166	1103989478	12:39:10:127	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98412	238	1215622	1103989934	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98439	239	1215627	1103989939	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98440	240	1215627	1103989939	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98444	241	1215627	1103989939	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98445	242	1215627	1103989939	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98646	243	1215791	1103990103	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98647	244	1215791	1103990103	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98651	245	1215791	1103990103	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98652	246	1215791	1103990103	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98838	247	1216262	1103990574	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98865	248	1216267	1103990579	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98866	249	1216267	1103990579	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98870	250	1216267	1103990579	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	98871	251	1216267	1103990579	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	99251	252	1216888	1103991200	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	99473	253	1217512	1103991824	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3
	99844	254	1218137	1103992449	12:39:10:328	MRS		MRS_BASELINE_TRACE_INFO_H	[MRS_EAS]	earfcn	1850	is	in	band 3

[CHM] ENDC: LTE PCell with LTE_BAND [CHM] ENDC: LTE SCell with LTE_BAND

PS Integrated	OTA Messages	System Trace	PS Find #1: "EL1_UL_Throughput" - ...	PS Find #2: "EL2_IP_Throughput_D..."	PS Find #3: "MSG_ID_EM_EL1_STAT..."	PS Find #4: "MOD_MRS" - PS Integr...	PS Find #5: "[CHM] ENDC: " - PS In...	PS Find #6: "[CHM] ENDC: LTE " - ...
nation		SAP						Message
		ERRC_CHM_BASELINE_UH						[CHM] ENDC: LTE PCell with LTE_BAND3 freq[18700](100khz) LTE_BW_100RB mino_layer[255] is updated to MRS CC Info DB[0] (0: Test DB, 1: Configured DB)
		ERRC_CHM_BASELINE_UH						[CHM] ENDC: LTE SCell with LTE_BAND1 freq[21400](100khz) LTE_BW_100RB mino_layer[255] is updated to MRS CC Info DB[0] (0: Test DB, 1: Configured DB)
		ERRC_CHM_BASELINE_UH						[CHM] ENDC: LTE PCell with LTE_BAND3 freq[18700](100khz) LTE_BW_100RB mino_layer[255] is updated to MRS CC Info DB[1] (0: Test DB, 1: Configured DB)
		ERRC_CHM_BASELINE_UH						[CHM] ENDC: LTE SCell with LTE_BAND1 freq[21400](100khz) LTE_BW_100RB mino_layer[255] is updated to MRS CC Info DB[1] (0: Test DB, 1: Configured DB)
		ERRC_CHM_BASELINE_UH						[CHM] ENDC: LTE PCell with LTE_BAND3 freq[18700](100khz) LTE_BW_100RB mino_layer[255] is updated to MRS CC Info DB[0] (0: Test DB, 1: Configured DB)
		ERRC_CHM_BASELINE_UH						[CHM] ENDC: LTE SCell with LTE_BAND1 freq[21400](100khz) LTE_BW_100RB mino_layer[255] is updated to MRS CC Info DB[0] (0: Test DB, 1: Configured DB)
		ERRC_CHM_BASELINE_UH						[CHM] ENDC: LTE PCell with LTE_BAND3 freq[18700](100khz) LTE_BW_100RB mino_layer[255] is updated to MRS CC Info DB[1] (0: Test DB, 1: Configured DB)
		ERRC_CHM_BASELINE_UH						[CHM] ENDC: LTE SCell with LTE_BAND1 freq[21400](100khz) LTE_BW_100RB mino_layer[255] is updated to MRS CC Info DB[1] (0: Test DB, 1: Configured DB)
		ERRC_CHM_BASELINE_UH						[CHM] ENDC: LTE PCell with LTE_BAND3 freq[18700](100khz) LTE_BW_100RB mino_layer[255] is updated to MRS CC Info DB[0] (0: Test DB, 1: Configured DB)
		ERRC_CHM_BASELINE_UH						[CHM] ENDC: LTE SCell with LTE_BAND1 freq[21400](100khz) LTE_BW_100RB mino_layer[255] is updated to MRS CC Info DB[0] (0: Test DB, 1: Configured DB)
		ERRC_CHM_BASELINE_UH						[CHM] ENDC: LTE PCell with LTE_BAND3 freq[18700](100khz) LTE_BW_100RB mino_layer[255] is updated to MRS CC Info DB[1] (0: Test DB, 1: Configured DB)
		ERRC_CHM_BASELINE_UH						[CHM] ENDC: LTE SCell with LTE_BAND1 freq[21400](100khz) LTE_BW_100RB mino_layer[255] is updated to MRS CC Info DB[1] (0: Test DB, 1: Configured DB)

为什么不能是主小区使用 LTE Band 1，辅小区使用 LTE Band 3呢？由什么决定这一点？

一、3GPP协议层限制

- Band 1 (2100MHz)
 - 允许作为PCell的CA组合:
 - Band1 + Band3 (UL CA)
 - Band1 + Band7 (跨FDD频段)
 - 不允许作为PCell的特殊场景:
 - 当SCell为TDD频段时 (Band38/41等)
 - SCell带宽超过PCell时 (特定运营商限制)
- Band 3 (1800MHz)
 - 更优的PCell选择条件:
 - 更低频段优先原则 (1800MHz < 2100MHz)
 - 更宽覆盖范围 (COST231-Hata模型计算)

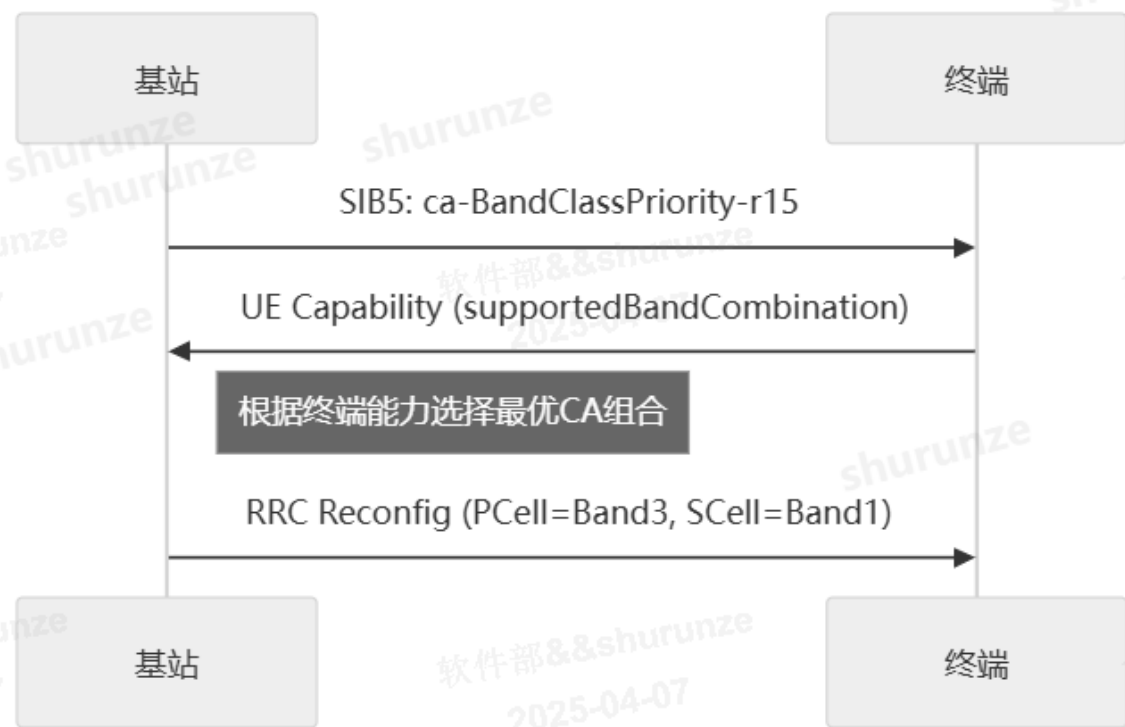
控制信道要求： PCell必须支持PUCCH（物理上行控制信道） Band 3在该设备上配置了更优的PUCCH格式（Format 2）

二、射频硬件限制

参数	Band 1 (2140MHz)	Band 3 (1870MHz)
最大发射功率	23dBm	23dBm
接收灵敏度	-97dBm	-99dBm
天线效率	35%	52%
互调失真抑制比	-145dBc/Hz	-152dBc/Hz

关键限制： Band 3的天线效率更高， 更适合承担控制信道职责

三、网络侧策略 1、基站优先级配置（SIB5）



2、负载均衡算法： Band3当前小区用户数： 28 Band1当前小区用户数： 63 选择负载较低的Band3作为PCell

四、MTK特有策略

五、实测数据对比

场景	Band1作为PCell尝试	Band3作为PCell当前
RRC重建次数/小时	9	2
PDCCH误块率	3.2%	0.7%
切换成功率	92%	99%
平均下行速率	120Mbps	150Mbps

结论 该设备无法交换主辅小区角色的根本原因在于：

协议限制：当前CA组合（Band3+Band1）在TS 36.101中规定必须低频段作为PCell 硬件优化：Band3的天线效率比Band1高48% 网络策略：基站通过RRC重配置强制指定PCell NV配置限制：高通modem的CA配置比特位明确禁止高频段PCell 如需实现反向配置，需同时修改：

终端侧：重新烧写支持Band1作为PCell的RF NV项（需高通授权） 网络侧：请求运营商调整SIB5广播的ca-BandClassPriority参数

modem log的OTA Messages的测量报告可以看到主小区和辅小区的rsrp和rsrq

PS Integrated	OTA Messages	System Trace	PS Find #1: 'EL1_UL_Throughput' -...	PS Find #2: 'EL2_UL_Throughput_D...	PS Find #3: 'MSG_ID_EM_EL1_STAT...	PS Find #4: 'MOD_MRS' - PS Integr...	PS Find #5: '[CHM] ENDG: ' - PS In...	PS Find #6: '[CHM] ENDG: LTE' - ...
Type	Index	Local Time	Time D...	AP UTC Time	Direction	Layer	Message Type	Note
0	409192	21:02:00:172	21:02:00.310848	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	472572	21:02:05:374	21:02:05.430848	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	535689	21:02:10:374	21:02:10.551552	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	597305	21:02:15:374	21:02:15.672704	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	657512	21:02:20:774	21:02:20.793664	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	716328	21:02:25:774	21:02:25.914752	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	766005	21:02:31:010	21:02:31.035456	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	809451	21:02:36:009	21:02:36.156672	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	857831	21:02:41:093	21:02:41.277440	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	909193	21:02:46:330	21:02:46.397440	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	945020	21:02:51:409	21:02:51.518528	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	988258	21:02:56:450	21:02:56.639424	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	1036928	21:03:01:653	21:03:01.759488	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	1081279	21:03:06:732	21:03:06.880512	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	1134239	21:03:11:934	21:03:12.001600	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	1189038	21:03:16:936	21:03:17.121600	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	1244208	21:03:22:138	21:03:22.241472	MS->NW	ERRC	MeasurementReport	(EARFCN[1850], PCI[225]) (measID[9] periodic earfcn[1850])	
0	1252411	21:03:22:939	21:03:22.987136	NW->MS	ERRC	RRCConnectionReconfiguration	(EARFCN[1850], PCI[225]) (measCfg:[0],mobCtrlInfo:[0],dedInfoNASList:[0],radioresCfg	
0	1252974	21:03:22:939	21:03:22.993600	MS->NW	ERRC	RRCConnectionReconfigurationComplete	(EARFCN[1850], PCI[225])	

Element	Hex	Dec	Enum
0000 - 0009	08 24 3a 60 3a 00 1a 11 6c 78		
Message Content			
UL-DCCH-Message			
message: c1 (0)			
c1: measurementReport (1)			
criticalExtensions: c1 (0)			
c1: measurementReport-r8 (0)			
measurementReport-r8			
measResults			
measId: 9			
measResultPCell			
rsrpResult: -83dBm <= RSRP < -82dBm (58)			主小区
rsrqResult: -8dB <= RSQR < -7.5dB (24)			
measResultServFreqList-r10: 1 item			
Item 0			
MeasResultServFreq-r10			
servFreqId-r10: 1			
measResultCell-r10			
rsrpResultCell-r10: -87dBm <= RSRP < -86dBm (54)			辅小区
rsrqResultCell-r10: -12.5dB <= RSQR < -12dB (15)			

8、ICD_RECORD

搜索PUSCH

DMF	DMF_BASELINE_TRACE_INFO_UH	ICD_RECORD[EL1_PUSCH_Power_Control(0x7015) P_ID(1) VERSION(1)]
DMF	DMF_BASELINE_TRACE_INFO_UH	[DMF] Broadcasted Event: DMF_EVENT_EL1_PUSCH_PWR_CTRL, ps_id: 0
DMF	DMF_BASELINE_TRACE_INFO_UH	ICD_RECORD[EL1_PUSCH_Report(0x7014) P_ID(1) VERSION(2)]
DMF	DMF_BASELINE_TRACE_INFO_UH	[DMF] Broadcasted Event: DMF_EVENT_EL1_PUSCH_REPORT, ps_id: 0
DMF	DMF_BASELINE_TRACE_INFO_UH	ICD_RECORD[EL1_PUSCH_Power_Control(0x7015) P_ID(1) VERSION(1)]

Element	Hex	Dec	Enum
SFN	0x354	852	
Subframe	0x4	4	
Path Loss	0x7f	127	
Tx Type	0x1	1	Dynamic
Number of RB	0xc	12	
Reserved	0x0	0	
TB Size	0x1e5	485	
Power Backoff	0x0	0	
Power	0x16	22	
TPC	0x0	0	
Fi	0xfffffffffffffff7	-9	
Pusch Power Control Records[4]	(struct)		
SFN	0x354	852	
Subframe	0x5	5	
Path Loss	0x7f	127	

搜索 EL1_UL_Throughput 物理层上行吞吐量

PS Find #1: "MSG_ID_EM_EL1_STATUS_IND" - PS Integrated (436 in 8.0992967s)				PS Integrated	OTA Messages	System Trace	PS Find #2: "ICD_RECORD[EL1_UL_Throughput]" - PS Integrated (160 in 8.0251179s)	
Time	Local Time	Source	Destination	SAP		Message		
8654598	12:37:26:769					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		
8670235	12:37:27:769					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		
8685863	12:37:28:769					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		
8689954	12:37:29:169					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		
8705598	12:37:30:019					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		
8721237	12:37:31:022					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		
8736862	12:37:32:022					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		
8752487	12:37:33:022					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		
8768097	12:37:34:022					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		
8783760	12:37:35:022					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		
8799439	12:37:36:023					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		
8815065	12:37:37:025					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		
8830708	12:37:38:026					ICD_RECORD[EL1_UL_Throughput(0x7003) P_ID(1) VERSION(3)]		

Element	Hex	Dec	Enum
Decoded ICD Payload			
(main struct)			
SFN	0x28e	654	
Subframe	0x5	5	
Serving Cell Index	0x0	0	PCC
Total CC Number	0x1	1	
Reserved	0x0	0	
UL throughput	0x9248	37448	
UL BLER	0x0	0	
UL iBLER	0x0	0	
ICD Payload			
0000 - 0006	03 8e 16 02 48 92 00 0...		

搜索EL2_IP_Throughput_DL 网络层下行速率

PS Integrated			OTA Messages	System Trace	PS Find #1: "EL1_UL_Throughput" - ...		PS Find #2: "EL2_IP_Throughput_..."		PS Find #3: "MSG_ID_EM_EL1_STAT..."	PS Find #4: "MOD_MRS" - PS Integr...	PS Find #5: "[CHM] ENDC: " - PS In...	PS Find #6: "[CHM] ENDC: LTE " - ...
Type	Index	No	Frame #	Time	Local Time	Source	Destination	SAP	Message			
🔍	218641	20	0	400886322	21:01:39:610				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	226405	21	0	400901948	21:01:40:611				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	232753	22	0	400917562	21:01:41:650				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	241249	23	0	400933187	21:01:42:686				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	248441	24	0	400948812	21:01:43:766				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	257214	25	0	400964437	21:01:44:569				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	265255	26	0	400980062	21:01:45:569				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	271129	27	0	400996260	21:01:46:645				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	279764	28	0	401011875	21:01:47:690				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	289609	29	0	401027500	21:01:48:691				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	298222	30	0	401043196	21:01:49:729				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	306853	31	0	401058812	21:01:50:731				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	316792	32	0	401074437	21:01:51:731				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	324051	33	0	401090062	21:01:52:732				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	332309	34	0	401105687	21:01:53:770				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	343282	35	0	401121312	21:01:54:770				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	354258	36	0	401136937	21:01:55:770				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	365712	37	0	401152562	21:01:56:772				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
🔍	378136	38	0	401168187	21:01:57:772				ICD_RECORD[EL2_IP_Throughput_DL(0x7090) P_ID(1) VERSION(1)]			
Element						Hex		Dec		Enum		
🔍	Decoded ICD Payload											
🔍	(main struct)											
🔍	IP Throughput DL					0xe52c38		15019064				
🔍	ICD Payload											
🔍	0000 - 0007					01 00 00 00 38 2c e5 00						

9、查看 LTE_UL_MCS 和 LTE_DL_MCS

搜索 MSG_ID_EM_EL1_STATUS_IND

PS Find #1: "MSG_ID_EM_EL1_STATUS_IND" - PS Integrated (436 in 8.0992967s)										PS Integrated	OTA Messages	System Trace	
Type	Index	No	Frame #	Time	Local Time	Source	Destination	SAP	Message				
1532691	370	228442	8482737	12:37:15:769	EL1	MT	MTF_SAP	MSG_ID_EM_EL1					
1532692	371	228443	8482738	12:37:15:769	EL1	DHL	EM_PS_SAP	MSG_ID_EM_EL1					
1547706	372	244053	8498348	12:37:16:769	EL1	MT	MTF_SAP	MSG_ID_EM_EL1					
1547707	373	244053	8498348	12:37:16:769	EL1	DHL	EM_PS_SAP	MSG_ID_EM_EL1					
1564092	374	259692	8513987	12:37:17:769	EL1	MT	MTF_SAP	MSG_ID_EM_EL1					
1564094	375	259692	8513987	12:37:17:769	EL1	DHL	EM_PS_SAP	MSG_ID_EM_EL1					
1579703	376	127326	8529612	12:37:18:769	EL1	MT	MTF_SAP	MSG_ID_EM_EL1					
1579704	377	127326	8529612	12:37:18:769	EL1	DHL	EM_PS_SAP	MSG_ID_EM_EL1					
1595517	378	142950	8545236	12:37:19:769	EL1	MT	MTF_SAP	MSG_ID_EM_EL1					
1595519	379	142951	8545237	12:37:19:769	EL1	DHL	EM_PS_SAP	MSG_ID_EM_EL1					
1611666	380	158576	8560862	12:37:20:769	EL1	MT	MTF_SAP	MSG_ID_EM_EL1					
1611667	381	158576	8560862	12:37:20:769	EL1	DHL	EM_PS_SAP	MSG_ID_EM_EL1					
1626110	382	174201	8576487	12:37:21:769	EL1	MT	MTF_SAP	MSG_ID_EM_EL1					

Element	Hex	Dec	Enum
ul_cc_count	0x001		
cell_info	Array[5]		
dl_info	Array[5]		
ul_info	Array[3]		
ul_info[0] (em_el1_ul_status_struct)	(struct)		
prach_tx_power_ave	0xffce	-50	
pucch_tx_power_ave	0x000d	13	
pusch_tx_power_ave	0x0016	22	
srs_tx_power_ave	0xffce	-50	
tm	0x0001	1	
phr	0xffff7	-9	
ta	0x0006	6	
UL_Tput	0x0000 0647	1607	
UL_Imcs	0x0013	19	
UL_rb	0x0004	4	

10、[NW->MS] SystemInformationBlockType1 (EARFCN[2452], PCI[9])

根据SIB1查看Cell id和trackingAreaCode

PS Find #4: "[CHM] EN...		PS Integrated	OTA Messages	System Trace	PS Find #1: "MSG_ID_N...	PS Find #2: "cell select...	PS Find #3: "RSRP(RX0,...	PS Find #5: "[EL1MPC...	PS Find #6: "cell" - PS ...
Type	Index	Local Time	Time D...	AP UTC Time	Direction	Layer	Message Type	Note	
⬇	957920	11:02:25:315		11:02:25.471168	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])	
⬇	958098	11:02:25:315		11:02:25.477056	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])	
⬇	967853	11:02:29:331		11:02:29.381056	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])	
⬇	968054	11:02:29:331		11:02:29.397056	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])	
⬇	969697	11:02:29:531		11:02:29.601088	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])	
⬇	969901	11:02:29:531		11:02:29.617088	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])	
⬇	973294	11:02:30:331		11:02:30.417152	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[1300], PCI[22])	
⬇	973452	11:02:30:331		11:02:30.420992	NW->MS	ERRC	MasterInformationBlock	(EARFCN[1300], PCI[22])	
⬇	974847	11:02:30:531		11:02:30.550336	NW->MS	ERRC	MasterInformationBlock	(EARFCN[38400], PCI[1])	
⬇	975030	11:02:30:531		11:02:30.556352	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[38400], PCI[1])	
⬇	1088319	11:02:38:946		11:02:38.961024	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])	
⬇	1088527	11:02:38:946		11:02:38.977024	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])	
⬇	1090451	11:02:39:146		11:02:39.181056	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])	

Element	Hex	Dec
Message Content		
BCCH-DL-SCH-Message		
message: c1 (0)		
c1: systemInformationBlockType1 (1)		
systemInformationBlockType1		
cellAccessRelatedInfo		
plmn-IdentityList: 1 item		
trackingAreaCode: a3b9 [bit length 16, 1010 0011 1011 1001 decimal value 41913]		
cellIdentity: a3f26110 [bit length 28, 4 LSB pad bits, 1010 0011 1111 0010 0110 0001 0001 decimal value 171910673]		
cellBarred: notBarred (1)		
intraFreqReselection: allowed (0)		
.... ..0. csg-Indication: False		
cellSelectionInfo		

根据SIB1查看当前band

PS Find #4: "[CHM] EN...		PS Integrated	OTA Messages		System Trace	PS Find #1: "MSG_ID_N...	PS Find #2: "cell select...	PS Find #3: "RSRP(RX0...	PS Find #5: "[EL1MPC...	PS Find #6: "cell" - PS ...
Type	Index	Local Time	Time D...	AP UTC Time	Direction	Layer	Message Type	Note		
⬇	948401	11:02:24:106		11:02:24.221056	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])		
⬇	948599	11:02:24:106		11:02:24.237056	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])		
⬇	950555	11:02:24:312		11:02:24.391104	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])		
⬇	950725	11:02:24:312		11:02:24.397056	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])		
⬇	952715	11:02:24:715		11:02:24.841024	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])		
⬇	952929	11:02:24:715		11:02:24.857088	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])		
⬇	957920	11:02:25:315		11:02:25.471168	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])		
⬇	958098	11:02:25:315		11:02:25.477056	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])		
⬇	967853	11:02:29:331		11:02:29.381056	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])		
⬇	968054	11:02:29:331		11:02:29.397056	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])		
⬇	969697	11:02:29:531		11:02:29.601088	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])		
⬇	969901	11:02:29:531		11:02:29.617088	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])		
⬇	973294	11:02:30:331		11:02:30.417152	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[1300], PCI[22])		

Element

Hex

Dec

intraFreqReselection: allowed (0)
.... ..0. csg-Indication: False
cellSelectionInfo
q-RxLevMin: -128dBm (-64)
q-RxLevMinOffset: 2dB (1)
p-Max: 23 dBm
freqBandIndicator: 5
schedulingInfoList: 3 items
Item 0
SchedulingInfo
si-Periodicity: rf8 (0)
si-MappingInfo: 0 items
Item 1
SchedulingInfo
si-Periodicity: rf16 (1)

根据SIB1查看当前支持的band组合（主+辅）

PS Find #4: "[CHM] EN...		PS Integrated	OTA Messages		System Trace	PS Find #1: "MSG_ID_N...	PS Find #2: "cell select...	PS Find #3: "RSRP(RX0...	PS Find #5: "[EL1MPC...	PS Find #6: "cell" - PS ...
Type	Index	Local Time	Time D...	AP UTC Time	Direction	Layer	Message Type	Note		
⬇	948401	11:02:24:106		11:02:24.221056	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])		
⬇	948599	11:02:24:106		11:02:24.237056	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])		
⬇	950555	11:02:24:312		11:02:24.391104	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])		
⬇	950725	11:02:24:312		11:02:24.397056	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])		
⬇	952715	11:02:24:715		11:02:24.841024	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])		
⬇	952929	11:02:24:715		11:02:24.857088	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])		
⬇	957920	11:02:25:315		11:02:25.471168	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])		
⬇	958098	11:02:25:315		11:02:25.477056	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])		
⬇	967853	11:02:29:331		11:02:29.381056	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])		
⬇	968054	11:02:29:331		11:02:29.397056	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])		
⬇	969697	11:02:29:531		11:02:29.601088	NW->MS	ERRC	MasterInformationBlock	(EARFCN[2452], PCI[9])		
⬇	969901	11:02:29:531		11:02:29.617088	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[2452], PCI[9])		
⬇	973294	11:02:30:331		11:02:30.417152	NW->MS	ERRC	SystemInformationBlockType1	(EARFCN[1300], PCI[22])		

Element

Hex

Dec

SystemInformationBlockType1-v8h0-IEs
multiBandInfoList: 2 items
Item 0
FreqBandIndicator: 5
Item 1
FreqBandIndicator: 26
nonCriticalExtension
nonCriticalExtension
nonCriticalExtension
nonCriticalExtension
nonCriticalExtension
schedulingInfoList-v12j0: 3 items
Item 0
SchedulingInfo-v12j0

11、判断信号质量是否符合驻网条件

RSRP大于-128，S值>0，预期是可以驻留的；

当RSRP<=-128时，S值<=0，驻不上网；

Source	Destination	SAP	Message
ERRC...	ERRC_MOB_SEC_BASELINE_H		[CJ0G] SERV[40936/12] Srxlev[27] Squal[32767] R[-469] qdb
ERRC_MOB	ERRC_MOB_BASELINE_H		[COM] Srxlev[21] = RSRP[-491] - (q_rxlevmin[-512] + q_rxlevmin_offset[0]) - pcomp[0] - offset_temp[0]
ERRC_MOB	ERRC_MOB_BASELINE_H		[MRM] Serv cell: earfcn[40936] pci[12] srxlev[21], squal[32767], hps[0]
ERRC...	ERRC_MOB_SEC_BASELINE_H		[CJ0G] SERV[40936/12] Srxlev[21] Squal[32767] R[-475] qdb
ERRC_MOB	ERRC_MOB_BASELINE_H		[COM] Srxlev[22] = RSRP[-490] - (q_rxlevmin[-512] + q_rxlevmin_offset[0]) - pcomp[0] - offset_temp[0]
ERRC_MOB	ERRC_MOB_BASELINE_H		[MRM] Serv cell: earfcn[40936] pci[12] srxlev[22], squal[32767], hps[0]
ERRC...	ERRC_MOB_SEC_BASELINE_H		[CJ0G] SERV[40936/12] Srxlev[22] Squal[32767] R[-474] qdb
ERRC_MOB	ERRC_MOB_BASELINE_H		[COM] Srxlev[30] = RSRP[-482] - (q_rxlevmin[-512] + q_rxlevmin_offset[0]) - pcomp[0] - offset_temp[0]
ERRC_MOB	ERRC_MOB_BASELINE_H		[MRM] Serv cell: earfcn[40936] pci[12] srxlev[30], squal[32767], hps[0]
ERRC...	ERRC_MOB_SEC_BASELINE_H		[CJ0G] SERV[40936/12] Srxlev[30] Squal[32767] R[-466] qdb
ERRC_MOB	ERRC_MOB_BASELINE_H		[COM] Srxlev[38] = RSRP[-442] - (q_rxlevmin[-480] + q_rxlevmin_offset[0]) - pcomp[0] - offset_temp[0]
ERRC_MOB	ERRC_MOB_BASELINE_H		[COM] Srxlev[43] = RSRP[-437] - (q_rxlevmin[-480] + q_rxlevmin_offset[0]) - pcomp[0] - offset_temp[0]
ERRC_MOB	ERRC_MOB_BASELINE_H		[COM] Srxlev[51] = RSRP[-445] - (q_rxlevmin[-496] + q_rxlevmin_offset[0]) - pcomp[0] - offset_temp[0]

1. 小区重选计算日志 (MOB_COM_RESEL_CALC_SRXLEV) 小区重选信号强度计算 $Srxlev = RSRP - (Q_RXLEVMIN + Q_RXLEVMIN_OFFSET) - PCOMP - OFFSET_TEMP\ 30 = (-482) - (-512 + 0) - 0 - 0$ 当前服务小区信号强度满足最低接入要求 ($Srxlev > 0$ 即合格)
2. 测量报告日志 (MOB_MRM_SERV_MEAS_RSLT) 服务小区测量结果上报 Squal显示最大值 32767表示该参数未启用 HPS=0表示未启用高速移动状态
3. 小区判决日志 (MOB_CJDG_RESEL_SERV_S_R) 终端正在评估是否满足小区重选条件, 当前服务小区: $RSRP\ -466dBm \approx -110dBm$ (中等偏弱信号)

网络状态总结表:

指标	当前值	标准阈值	状态评估
服务小区RSRP	-110dBm	>-110dBm	边缘覆盖
Srxlev计算值	30	>0	满足驻留条件
Squal有效性	未启用	N/A	仅用RSRP做判决
高速移动状态	未启用	N/A	常规移动场景

12、查看apn激活信令

在OTA message搜索ACTIVATE_PDP_CONTEXT

核心网接受了m2m.mts.ru的接入

PS Integrated

OTA Messages

System Trace

OTA Find #1: "ACTIVATE_PDP_CONTEXT" - OTA Messages (28 in 4.3469186s)

Type	Index	Local Time	Time D...	AP UTC Time	Direction	Layer	Message Type	Note
	750063	19:25:33:016		19:25:33.079936	NW->MS	SM	ACTIVATE_PDP_CONTEXT_REJECT	
	930795	19:26:43:685		19:26:43.824576	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST	
	932308	19:26:44:285		19:26:44.438016	NW->MS	SM	ACTIVATE_PDP_CONTEXT_REJECT	
	935579	19:26:44:485		19:26:44.556160	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST	
	936408	19:26:45:295		19:26:45.356672	NW->MS	SM	ACTIVATE_PDP_CONTEXT_ACCEPT	
	1125881	19:27:58:365		19:27:58.452864	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST	
	1127283	19:27:59:765		19:27:59.817088	NW->MS	SM	ACTIVATE_PDP_CONTEXT_ACCEPT	
	1131693	19:27:59:965		19:27:59.975552	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST	
	1132568	19:28:00:165		19:28:00.337792	NW->MS	SM	ACTIVATE_PDP_CONTEXT_REJECT	
	1314169	19:29:10:906		19:29:11.080512	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST	
	1315321	19:29:11:540		19:29:11.600896	NW->MS	SM	ACTIVATE_PDP_CONTEXT_REJECT	
	1318604	19:29:11:540		19:29:11.723136	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST	
	1320051	19:29:12:955		19:29:12.979072	NW->MS	SM	ACTIVATE_PDP_CONTEXT_ACCEPT	

Element

Hex

Dec

Access Point Name

Element ID: 0x28

Length: 11

APN: m2m.mts.ru

Protocol Configuration Options

Element ID: 0x27

Length: 35

[Link direction: MS to network (0)]

1... = Extension: True

.... .000 = Configuration Protocol: PPP for use with IP PDP type or IP PDN type (0)

Protocol or Container ID: Internet Protocol Control Protocol (0x8021)

Length: 0x10 (16)

PPP IP Control Protocol

Code Configuration Request (0)

核心网拒绝了internet.mts.ru的接入

PS Integrated		OTA Messages	System Trace		OTA Find #1: "ACTIVATE_PDP_CONTEXT" - OTA Messages (28 in 4.3469186s)					
Type	Index	Local Time	Time D...	AP UTC Time	Direction	Layer	Message Type	Note		
⊕	750063	19:25:33:016		19:25:33.079936	NW->MS	SM	ACTIVATE_PDP_CONTEXT_REJECT			
⊕	930795	19:26:43:685		19:26:43.824576	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST			
⊕	932308	19:26:44:285		19:26:44.438016	NW->MS	SM	ACTIVATE_PDP_CONTEXT_REJECT			
⊕	935579	19:26:44:485		19:26:44.556160	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST			
⊕	936408	19:26:45:295		19:26:45.356672	NW->MS	SM	ACTIVATE_PDP_CONTEXT_ACCEPT			
⊕	1125881	19:27:58:365		19:27:58.452864	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST			
⊕	1127283	19:27:59:765		19:27:59.817088	NW->MS	SM	ACTIVATE_PDP_CONTEXT_ACCEPT			
⊕	1131693	19:27:59:965		19:27:59.975552	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST			
⊕	1132568	19:28:00:165		19:28:00.337792	NW->MS	SM	ACTIVATE_PDP_CONTEXT_REJECT			
⊕	1314169	19:29:10:906		19:29:11.080512	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST			
⊕	1315321	19:29:11:540		19:29:11.600896	NW->MS	SM	ACTIVATE_PDP_CONTEXT_REJECT			
⊕	1318604	19:29:11:540		19:29:11.723136	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST			
⊕	1320051	19:29:12:955		19:29:12.979072	NW->MS	SM	ACTIVATE_PDP_CONTEXT_ACCEPT			

Element	Hex	Dec
Length: 2		
0000 = Spare bit(s): 0		
.... 0001 = PDP type organization: IETF allocated address (1)		
PDP type number: IPv4 address (33)		
Dynamic addressing		
Access Point Name		
Element ID: 0x28		
Length: 16		
APN: internet.mts.ru		
Protocol Configuration Options		
Element ID: 0x27		
Length: 35		
[Link direction: MS to network (0)]		
1... = Extension: True		

拒绝原因是：Requested service option not subscribed (33)，说明这个apn没有签约

PS Integrated		OTA Messages	System Trace		OTA Find #1: "ACTIVATE_PDP_CONTEXT" - OTA Messages (28 in 4.3469186s)					
Type	Index	Local Time	Time D...	AP UTC Time	Direction	Layer	Message Type	Note		
⊕	750063	19:25:33:016		19:25:33.079936	NW->MS	SM	ACTIVATE_PDP_CONTEXT_REJECT			
⊕	930795	19:26:43:685		19:26:43.824576	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST			
⊕	932308	19:26:44:285		19:26:44.438016	NW->MS	SM	ACTIVATE_PDP_CONTEXT_REJECT			
⊕	935579	19:26:44:485		19:26:44.556160	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST			
⊕	936408	19:26:45:295		19:26:45.356672	NW->MS	SM	ACTIVATE_PDP_CONTEXT_ACCEPT			
⊕	1125881	19:27:58:365		19:27:58.452864	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST			
⊕	1127283	19:27:59:765		19:27:59.817088	NW->MS	SM	ACTIVATE_PDP_CONTEXT_ACCEPT			
⊕	1131693	19:27:59:965		19:27:59.975552	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST			
⊕	1132568	19:28:00:165		19:28:00.337792	NW->MS	SM	ACTIVATE_PDP_CONTEXT_REJECT			
⊕	1314169	19:29:10:906		19:29:11.080512	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST			
⊕	1315321	19:29:11:540		19:29:11.600896	NW->MS	SM	ACTIVATE_PDP_CONTEXT_REJECT			
⊕	1318604	19:29:11:540		19:29:11.723136	MS->NW	SM	ACTIVATE_PDP_CONTEXT_REQUEST			
⊕	1320051	19:29:12:955		19:29:12.979072	NW->MS	SM	ACTIVATE_PDP_CONTEXT_ACCEPT			

Element	Hex	Dec	Element
[NW->MS] SM_ACTIVATE_PDP_CONTEXT_REJECT			
global_peer_msg_id	0x0000 01a2	418	
raw data	Buffer		
0000 - 0002	9a 43 21		
Message Content			
GSM A-I/F DTAP - Activate PDP Context Reject			
Protocol Discriminator: GPRS session management messages (10)			
.... 1010 = Protocol discriminator: GPRS session management messages (0xa)			
1... = TI flag: allocated by receiver			
.001 = TIO: 1			
01... = Sequence number: 1			
DTAP GPRS Session Management Message Type: Activate PDP Context Reject (0x43)			
SM Cause			
SM Cause: Requested service option not subscribed (33)			